In-Class Practice: Nonlinear PK

Module 8 - Nonlinear Pharmacokinetics · In-class activity · 14 Oct 2026

Name _____ Date _____

Question 1 What daily dose rate of a non-linear drug would provide a steady-state concentration of 15 mg/L? In this patient, the KM is 6.9 mg/L and the Vmax is 584 mg/day.

Question 2 When a 25 year-old, 75 kg patient was taking a daily oral dose of 350 mg, his steady state plasma concentration was 9.2 mg/L. When the daily dose was increased to 450 mg, a steady-state plasma concentration of 20 mg/L resulted. What is the KM of this drug in this patient?

Question 3 2. When a 25 year-old, 75 kg patient was taking a daily oral dose of 350 mg, his steady state plasma concentration was 9.2 mg/L. When the daily dose was increased to 450 mg, a steady-state plasma concentration of 20 mg/L resulted. 3. Based on the KM found in the previous question, what is the patient's Vmax?

4a 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) a. Calculate the patient's KM

4b 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) b. Calculate the patient's Vmax

4c 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) c. Recommend a daily dose of phenytoin for this patient to achieve a phenytoin plasma concentration of 15 mg/L at steady state.

4d 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) d. Calculate the half-life and the total body clearance when the plasma concentration is 15 mg/L. What is the half-life?

4d 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) d. Calculate the half-life and the total body clearance when the plasma concentration is 15 mg/L. What is the total body clearance?

4e 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) e. What steady-state concentration should be achieved if the dose were 400 mg daily?